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## Research Scientist in Artificial Intelligence

## Summary


I am a Research Scientist in AI within the Centre Inria d'Université Côte d'Azur, member of the Wimmics research team. Previously, I was a Junior Fellow in AI at Université Côte d'Azur (2023 – 2025), and a researcher in Artificial Intelligence in the industry, with Orange (2020 – 2023). I received my Doctoral degree in Computer Science from Université de Lorraine, Nancy, France, in 2020.
My research focuses on neuro-symbolic AI with and for knowledge graphs. I investigate interactions between domain knowledge in knowledge graphs and different forms of reasoning in a neuro-symbolic perspective (*e.g.*, injection of domain knowledge in Machine Learning models, analogical reasoning). In particular, I study such interactions in the context of the lifecycle of knowledge graphs (construction, matching, refinement, mining, knowledge discovery), and their usage in downstream applications (*e.g.*, recommender systems, explainable AI). My work involves both theoretical and applied perspectives, often in interdisciplinary settings (*e.g.*, biomedical, educational domains). I was the Principal Investigator of the ECLADATTA French national project and I am an Investigator of the AT2TA French national project (both funded by the French National Research Agency).


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## Current position

Since Oct. 2025 **Research Scientist – Centre Inria d'Université Côte d'Azur – Sophia Antipolis, France**
Research within the Wimmics team, common with the I3S laboratory (Université Côte d'Azur, CNRS)
Teaching within EFELIA Côte d'Azur and Polytech Nice Sophia ($\sim$ 64h of teaching)

## Research stays

Feb.-Jul. 2025 **Visiting Fellow – Télécom Paris – Paris, France**
Regular visits with Mehwish Alam (5 weeks in total)
Grant of 14.4k€ as part of the SHACKLE project

Jan. 2025 **Visiting Fellow – Università degli Studi di Bari Aldo Moro – Bari, Italy**
3-week visit with Claudia d'Amato
Grant of 4k€ from Università degli Studi di Bari Aldo Moro

Nov. 2024 **Visiting Fellow – INESC-ID, Instituto Superior Técnico, U. de Lisboa – Lisbon, Portugal**
2-week visit with Miguel Couceiro

Jan. 2024 **Visiting Fellow – Università degli Studi di Bari Aldo Moro – Bari, Italy**
3-week visit with Claudia d'Amato
Grant of 4k€ from the I3S laboratory

## Education

2016 – 2020 **PhD in Computer Science – Université de Lorraine – Nancy, France**

Matching and mining in knowledge graphs of the Web of data
Applications in pharmacogenomics

- Knowledge matching: identification of identical, more specific, or related knowledge units
  - Rule-based approach with ontology-based preorders (ICCS 2020 [45])
  - Graph embedding and clustering-based approach with Graph Convolutional Networks (DL4KG@ESWC 2019 [5], Semantic Web Journal [33])
- Knowledge mining
  - Scalable mining of path patterns (ALGOS 2020 [19])
  - Extension of Formal Concept Analysis to refine ontology class hierarchies (ISMIS 2017 [20], BDA 2017 (poster) [54], FCA4AI@IJCAI-ECAI 2018 [35])
- Explainable AI
  - Interpretable features and models to reproduce expert classifications of drugs causing or not side effects (post-hoc explanations) (BMC Medical Informatics and Decision Making [6])
- Throughout all works: importance of domain knowledge represented in ontologies
- Applications on real data in pharmacogenomics: more than 50k $n$-ary relationships linking sets of drugs, genetic factors, and phenotypes, from heterogeneous sources (different vocabularies, levels of granularity)
- Construction and maintenance of PGxLOD, a knowledge graph about pharmacogenomics (BMC Bioinformatics [7])
- Emphasis on Open Science
  - Codes on GitHub
  - Public knowledge graph and ontology
  - Deployability and reproducibility with Docker containers

Keywords: Ontology, $n$-ary Tuple, Preorder, Graph Embedding, Path Pattern, Formal Concept Analysis

Within the PractiKPharma project funded by the French National Research Agency (ANR)

Thesis supervisors :

- Amedeo Napoli (Senior Researcher, CNRS, LORIA)
- Adrien Coulet (Associate Professor, Université de Lorraine, LORIA)

Thesis written and defended in English on December 16th, 2020 with the following committee:

| | | |
|---|---|---|
| Chair | Marianne Clausel | Full Professor, Université de Lorraine, IECL |
| Reviewers | Claudia d'Amato | Associate Professor, University of Bari |
| | Fabien Gandon | Senior Researcher, Inria Sophia Antipolis - Méditerranée |
| Examiners | Céline Rouveirol | Full Professor, Université Sorbonne Paris Nord |
| | Matthias Samwald | Associate Professor, Medical University of Vienna |

2012 – 2015 **Engineering Diploma (M.Sc. equivalent) – TELECOM Nancy – Nancy, France**
Major in Software engineering – Valedictorian

Engineering master thesis
*3D visualization applied to aircraft MRO.*
*What purposes? What technologies? What developments?*
Based on my internship at Swiss AviationSoftware – Graded 17/20

2010 – 2012 **Integrated preparatory class – ECE – Paris, France**
Valedictorian

2010 **Scientific Baccalaureate – Lycée Stanislas – Villers-Lès-Nancy, France**
Major in Mathematics – with distinction

## Previous professional positions

2023 – 2025 **Junior Fellow in AI – Université Côte d'Azur – Sophia Antipolis, France**
Research within the Wimmics team of the I3S laboratory
Teaching within EFELIA Côte d'Azur and Polytech Nice Sophia ($\sim$ 128h of teaching)

2023 – 2024 **Temporary lecturer – TELECOM Nancy – Nancy, France**
18-hour class about artificial intelligence

2020 – 2023 **Researcher – Orange – Belfort, France**
Permanent contract
Knowledge Engineering and Machine Learning for knowledge graph construction and refinement

2022 – 2023 **Temporary lecturer – TELECOM Nancy – Nancy, France**
8-hour class about knowledge-based systems

2016 – 2020 **PhD student in Computer Science – LORIA / Université de Lorraine – Nancy, France**

- 2019 – 2020 Temporary teaching and research fellow (ATER) – IDMC
  $\sim$ 192h of teaching
- 2016 – 2019 Teaching assistant – TELECOM Nancy
  $\sim$ 64h of teaching per year

2015 – 2016 **Temporary lecturer – TELECOM Nancy – Nancy, France**
$\sim$14h of teaching

2015 – 2016 **Software developer – Swiss AviationSoftware – R&D department – Basel, Switzerland**
6-month internship; followed by a permanent contract
Development of a UI component to display animated 3D models from aircraft documentation

2010 – 2015 **Private teacher – Student tutoring associations**
Private lessons or groups of 10 to 30 students
Mathematics, programming, compiler theory, signal processing

2014 **Student Entrepreneurship Department (PeeL) – Université de Lorraine – Nancy, France**
3-month internship; entrepreneurial project

2013 **Caisse d'Epargne – Human Resources Information System – Metz, France**
3-month fixed-term contract
Development of a software analyzing HR databases to automatically produce organization charts (divisions, positions, people)

2012 **Caisse d'Epargne – Human Resources Information System – Metz, France**
9-week internship
Agile development of a software for career management within a 3-person team

## Awards

- **Best paper award [49]** at the French Conference IC ("Ingénierie des connaissances" - Knowledge Engineering) 2026 co-located with PFIA 2026
- **Best research paper award [11]** at the 23rd European Semantic Web Conference (ESWC 2026)
- **Best student research paper award [13]** at the 21st European Semantic Web Conference (ESWC 2024)
- **Best resource paper award [14]** at the 21st European Semantic Web Conference (ESWC 2024)
- **Best paper award [18]** at the 23rd International Conference on Knowledge Engineering and Knowledge Management (EKAW 2022)
- **Best paper award [51]** at the French Conference IC ("Ingénierie des connaissances" - Knowledge Engineering) 2022 co-located with PFIA 2022
- **Best thesis award [60]** at French Conference EGC 2022, awarded by the French Association EGC ("Extraction et Gestion des Connaissances" - Knowledge Extraction and Engineering)
- **1st prize [31]** (Accuracy Track) at the Semantic Web Challenge on Tabular Data to Knowledge Graph Matching 2021, co-located with the 20th International Semantic Web Conference (ISWC 2021)
- **3rd prize [32]** at the Semantic Web Challenge on Tabular Data to Knowledge Graph Matching 2020, co-located with the 19th International Semantic Web Conference (ISWC 2020)

## Funding of research projects

Attractivity Package **Principal Investigator** 2026 – 2029
Competitive funding awarded to newly recruited researchers through a project-based selection process, within the Initiative Of Excellence Université Côte d'Azur

This project supports my research activities on neuro-symbolic AI with and for knowledge graphs.

Funding for my institution (Inria): 50k€
Total project funding: 50k€

SHACKLE **Principal Investigator** Feb.-Jul. 2025
Sub-grant agreement of ENFIELD, a EU Horizon Research and Innovation Action

This project funds regular research stays to study the neuro-symbolic integration of validation schemata (*i.e.* SHACL shapes) from knowledge graphs in Machine Learning models to improve their soundness and completeness.

Funding for my institution (Université Côte d'Azur): 14.4k€
Total project funding: 14.4k€
Partners: Télécom Paris, Université Côte d'Azur (PI)

ECLADATTA **Principal Investigator (2023); Associate Member (2023 – 2026)** 2023 – 2026
French National Research Agency (ANR) Generic Call 2022

I set up and led during 6 months this 42-month project which aims at proposing a joint pipeline for extracting and reconciling knowledge from texts, tables, and knowledge graphs. After leaving my position at Orange, I became an associate member.

Funding for my institution (Orange): 196k€
Total project funding: 601k€
Partners: EURECOM, IRIT, Orange (PI)

AT2TA — **Investigator; WP co-leader** — 2023 – 2026
French National Research Agency (ANR) Generic Call 2022

This 42-month project aims at exploring the theory and applications of analogical reasoning. I am co-leading the work package on use cases and applications. I am leading the task studying interactions between knowledge management and analogical reasoning. After leaving my position at Orange, I transferred this project to Université Côte d'Azur.

Funding for my institution (Orange then Université Côte d'Azur): 30k€
Total project funding: 669k€
Partners: LORIA (PI), IHU Imagine, IRIT, Inria, Infologic, Université Côte d'Azur

Participant in: KG-TWIN (PEPR ENSEMBLE, 2026 – 2030), SAFE-KG (ANR PRC, 2026 – 2030), MetabolinkAI (ANR-SNF PRCI, 2025 – 2029)

## Community services

### Event organization activities

| | |
|---|---|
| PGx Track 2026 | Co-organizer with Adrien Coulet<br>Within the Ontology Alignment Evaluation Initiative 2026 (OAEI 2026) |
| ESWC 2026 | Web & Publicity co-chair with Tabea Tietz and Ignacio García<br>23rd European Semantic Web Conference<br>∼ 500 attendees |
| Course at URS 2025 | Co-organizer with Miguel Couceiro, Ndeye-Emilie Mbengue, and Lucas Jarnac<br>Course and shared-task "Analogical reasoning through the lens of KGs"<br>Within the Unite! Research School 2025<br>Curriculum preparation and 1-hour class on "Knowledge Graphs & Machine Learning"<br>∼ 20 attendees |
| PGx Track 2025 | Co-organizer with Adrien Coulet<br>Within the Ontology Alignment Evaluation Initiative 2025 (OAEI 2025) |
| Principia Analogiae 2025 | Co-general chair with Miguel Couceiro, Dafna Shahaf, and Zied Bouraoui<br>∼ 20 attendees |
| 3rd M4D2 | Co-organizer with Miguel Couceiro<br>3rd Symposium Mathematics for Decision and Discovery<br>∼ 20 attendees |
| PGx Track 2024 | Co-organizer with Adrien Coulet<br>Within the Ontology Alignment Evaluation Initiative 2024 (OAEI 2024) |
| Neuro-symbolic AI 2023 | Co-general chair with Zied Bouraoui and Fatiha Saïs<br>International workshop supported by AfIA and GDR RADIA<br>Co-located with the international conference ECSQARU 2023<br>∼ 80 attendees |
| PGx Track 2023 | Co-organizer with Adrien Coulet<br>Within the Ontology Alignment Evaluation Initiative 2023 (OAEI 2023) |
| ATA 2023 | General co-chair with Miguel Couceiro, Fadi Badra, and Esteban Marquer<br>International workshop on Analogies: from Theory to Applications<br>Co-located with the international conference ICCBR 2023<br>∼ 25 attendees |
| ISWC 2022 | Proceedings & Metadata co-chair with Giuseppe Pirrò<br>21st International Semantic Web Conference<br>∼ 500 attendees |

| | |
|---|---|
| ATA 2022 | General co-chair with Miguel Couceiro, Esteban Marquer, and Pierre-Alexandre Murena<br>International workshop on Analogies: from Theory to Applications<br>Co-located with the international conference ICCBR 2022<br>$\sim$ 25 attendees |
| ALGOS 2020 | General co-chair with Miguel Couceiro and Amedeo Napoli<br>International conference on ALgebras, Graphs and Ordered Sets<br>$\sim$ 200 attendees |

## Reviewing activities

- International Journals
  - Transactions on Graph Data and Knowledge (Editorial Board member)
  - Special Issue "Analogies: from Mathematical Foundations to Applications and Interactions with ML and AI" in Annals of Mathematics and Artificial Intelligence (Guest Editor)
  - Special Issue in honor of Maurice Pouzet in Discrete Mathematics & Theoretical Computer Science (Guest Editor)
  - Special Issue on Neurosymbolic Benchmark Papers of the Neurosymbolic Artificial Intelligence Journal (Guest Editorial Board member)
  - Special Issue on Neurosymbolic AI and Ontologies of the Neurosymbolic Artificial Intelligence Journal (Guest Editorial Board member)
  - Special Issue on Tools & Systems of the Semantic Web Journal (Guest Editorial Board member)
  - Special Issue on Ontology Matching and Machine Learning of the Semantic Web Journal (Guest Editorial Board member)
  - Semantic Web Journal
  - Journal of Web Semantics
  - IEEE Transactions on Knowledge and Data Engineering
  - International Journal of Approximate Reasoning
  - Data & Knowledge Engineering
  - The European Journal on Artificial Intelligence
  - Discrete Mathematics & Theoretical Computer Science
  - Communications Medicine
  - SoftwareX
  - Neurocomputing
  - Scientific Reports
  - Pattern Recognition Letters
  - ACM Computing Surveys
  - Engineering Applications of Artificial Intelligence
- International Conferences
  - TheWebConf 2027 (Semantics & Knowledge Track Program Committee member)
  - AAAI 2027 (Program Committee member)
  - EKAW 2026 (Program Committee member)
  - KR 2026 (KR meets Machine Learning and Explanation Track Program Committee member)
  - CIKM 2026 (Program Committee member)
  - ISWC 2026 (Posters & Demos Track Program Committee member)
  - ISWC 2026 (Research Track subreviewer)
  - ECML-PKDD 2026 (Research Track Program Committee member)
  - ESWC 2026 (Resources Track Program Committee member)
  - SAC 2026 (Knowledge & Natural Language Processing Track Program Committee member)
  - TheWebConf 2026 (Semantics & Knowledge Track Program Committee member)

  - AAAI 2026 (Program Committee member)
  - ISWC 2025 (Posters & Demos Track Program Committee member)
  - ECAI 2025 (Program Committee member)
  - ECML-PKDD 2025 (Research Track Program Committee member)
  - ESWC 2025 (Resources Track Program Committee member)
  - SAC 2025 (Knowledge & Natural Language Processing Track Program Committee member)
  - TheWebConf 2025 (Semantics & Knowledge Track Program Committee member)
  - MedInfo 2025 (reviewer)
  - EKAW 2024 (Program Committee member)
  - ISWC 2024 (Posters & Demos Track Program Committee member)
  - CIKM 2024 (additional reviewer)
  - ECML-PKDD 2024 (Research Track Program Committee member)
  - ECAI 2024 (Program Committee member)
  - ESWC 2024 (Resources Track Program Committee member)
  - TheWebConf 2024 (Program Committee member)
  - ISWC 2023 (Research Track Program Committee member)
  - ECML-PKDD 2023 (Research Track Program Committee member)
  - EKAW 2022 (additional reviewer)
  - MedInfo 2021 (reviewer)
  - IJCAI 2019 (assistant reviewer)
- International Workshops
  - XAI-KRKG, co-located with ECAI 2025 (Program Committee member)
  - XAI-KG, co-located with ESWC 2025 (Program Committee member)
  - AIMLAI, co-located with ECML-PKDD 2024 (subreviewer)
  - DL4KG, co-located with ISWC 2023 (Program Committee member)
  - AIMLAI, co-located with ECML-PKDD 2023 (subreviewer)
  - IARML, co-located with IJCAI 2023 (Program Committee member)
  - DL4KG, co-located with ISWC 2022 (Program Committee member)
  - IARML, co-located with IJCAI-ECAI 2022 (Program Committee member)
- National Conferences
  - EGC 2027 (Program Committee member)
  - IC 2026 (Program Committee member)
  - EGC 2026 (Program Committee member)
  - RJCIA 2024 (Program Committee member)
- Funding agencies
  - ENFIELD EU project – 2nd Innovation Scheme Open Call (2025)
  - Région Normandie – WINNINGNormandy Fellowship Program (2022)
- PhD theses
  - Raffaele Scaringi – Università degli Studi di Bari Aldo Moro – Bari, Italy (2025, reviewer)
- Monitoring Committees of PhD theses
  - Sayf Eddine Halmi - Université Côte d'Azur (from 2025)

### Miscellaneous

- Spokesperson (2025 – 2027) of AfIA ("Association française pour l'Intelligence Artificielle" – French Artificial Intelligence Society)
- Member of the GOBLIN EU COST ACTION, leader of Task 3.1 "KGs and Deep Learning"
- Member of the Cognitive AI W3C Community Group
- Member of GDR RADIA ("Groupement De Recherche Raisonnement, Apprentissage et Décision en Intelligence Artificielle" – Research Group on Reasoning, Learning, and Decision in Artificial Intelligence)
- Member of SIF ("Société informatique de France" – French Computer Science Society)

## Supervision

### PhD students (9)

Oct. 2026 – Oct. 2029 **Jovan Vukovic – Université Côte d'Azur**
Computational approaches for knowledge graph mining and completion dealing with uncertainty (MetabolinkAI project)
Co-supervised with Catherine Faron and Andrea Tettamanzi

Feb. 2026 – Feb. 2029 **Krysto Dagues de la Hellerie – Université Côte d'Azur / Berger-Levrault (CIFRE)**
Representing intervention scenarios and decisions using knowledge graphs: application to industrial maintenance
Co-supervised with Fabien Gandon, Catherine Faron, and Mokhtar Billami

Dec. 2025 – Dec. 2028 **Hanna Abi Akl – Université Côte d'Azur / DSTI**
Neuro-symbolic and knowledge-based approaches to develop reasoning abilities in Large Language Models
Co-supervised with Fabien Gandon and Catherine Faron
Publications: [8, 21]

Sept. 2025 – Sept. 2028 **Génesis Montenegro – Université Côte d'Azur / Berger-Levrault (CIFRE)**
From raw corpus to lexical-semantic graph: a methodological framework for industrial, agile and high-performance GraphRAG
Co-supervised with Fabien Gandon, Catherine Faron, and Mokhtar Billami
Publication: [39]

Oct. 2025 – Oct. 2028 **Ndeye-Emilie Mbengue – Université Côte d'Azur / Universidade de Lisboa**
Abstraction and analogical reasoning to improve knowledge representation in low-resource languages
Co-supervised with Fabien Gandon and Miguel Couceiro
Publication: [40]

Oct. 2024 – Oct. 2027 **Guillaume Méroué – Université Côte d'Azur**
Decentralized coordination of intelligent processes to improve the quality of a collection of knowledge graphs
Co-supervised with Fabien Gandon
Publication: [11, 49]

Oct. 2024 – Oct. 2027 **Nicolas Robert – Université Côte d'Azur**
Knowledge graph embedding models: symbolic knowledge injection and discovery
Co-supervised with Catherine Faron
Publication: [23, 37, 38]

Mar. 2023 – Mar. 2026 **Lucas Jarnac – Orange / Université de Lorraine (CIFRE)**
Reconciliation of uncertain knowledge from heterogeneous sources for knowledge graph refinement
Co-supervised with Miguel Couceiro
I supervised Lucas Jarnac until the end of my contract with Orange (August 2023)
Publications: [17, 41, 47, 57]

Sept. 2021 – Sept. 2024 **Nicolas Hubert – Université de Lorraine**
From semantic-aware to semantic-driven knowledge graph embedding models for link prediction
Without being his official supervisor, I collaborated with Nicolas and his supervisors (Armelle Brun and Davy Monticolo) on his thesis.
Publications: [3, 13, 14, 16, 18, 28, 55, 56, 70]

## Interns and student projects (26)

Apr. 2026 – Sept. 2026 **Thomas Santos, M.Sc. Year 2 (Bac+5)**
6-month internship
Analogical reasoning for KG management tasks: Entity Alignment and GraphRAG
Co-supervised with Fabien Gandon and Ndeye-Emilie Mbengue

Mar. 2026 – Aug. 2026 **Yaël Burdese, M.Sc. Year 2 (Bac+5)**
6-month internship
Comprehensively comparing GraphRAG approaches: from constructed graphs to generated results
Co-supervised with Fabien Gandon and Catherine Faron

Mar. 2026 – Aug. 2026 **Mattia Sernia, M.Sc. Year 2 (Bac+5)**
6-month internship
Knowledge graphs as a structured memory for collaborative agents
Co-supervised with Fabien Gandon, in collaboration with the Forgeron3 company

Mar. 2026 – Aug. 2026 **Alban Falcoz, M.Sc. Year 2 (Bac+5)**
6-month internship
Exploring the Invariance of Analogical Reasoning Across Different Knowledge Graph Embedding Models
Co-supervised with Fabien Gandon and Ndeye-Emilie Mbengue

Mar. 2026 – Aug. 2026 **Nadjib Zahaf Boualem, M.Sc. Year 2 (Bac+5)**
6-month internship
Entity alignment in knowledge graphs, application to pharmacogenomic knowledge
Co-supervised with Joël Legrand

Oct. 2025 – Mar. 2026 **Yaël Burdese, M.Sc. Year 2 (Bac+5)**
6-month research project
Comprehensively comparing GraphRAG approaches: from constructed graphs to generated results
Co-supervised with Fabien Gandon and Catherine Faron

Mar. 2025 – Aug. 2025 **Krysto Dagues de la Hellerie, M.Sc. Year 2 (Bac+5)**
6-month internship
Semantic Web Retrieval Augmented Generation
Co-supervised with Fabien Gandon, in collaboration with the Forgeron3 company

Apr. 2024 – Aug. 2024 **Adrien Chassaing-Monjou, M.Sc. Year 2 (Bac+5)**
5-month internship
Analogical Reasoning for Knowledge Graph Refinement
Co-supervised with Miguel Couceiro

Mar. 2024 – Jul. 2024 **Guillaume Méroué, M.Sc. Year 2 (Bac+5)**
4-month internship
Analogical Reasoning for Link Prediction in Knowledge Graphs
Co-supervised with Fabien Gandon

Oct. 2023 – Mar. 2024 **Lynda Attouche, M.Sc. Year 2 (Bac+5)**
6-month research project
Improving automatic text classification by leveraging class hierarchies
Co-supervised with Catherine Faron and Molka Dhouib

Oct. 2023 – Mar. 2024 **Guillaume Méroué, M.Sc. Year 2 (Bac+5)**
6-month research project

Analogical Reasoning for Link Prediction in Knowledge Graphs
Co-supervised with Miguel Couceiro

Oct. 2023 – Jun. 2024 **Mohammad Al Takach, Abbas Awarkeh, Bassem Jaber, M.Sc. Year 1 (Bac+4)**
8-month research project
Analogical Reasoning for Link Prediction in Knowledge Graphs
Co-supervised with Miguel Couceiro

Jun. 2023 – Aug. 2023 **Adrien Chassaing-Monjou, M.Sc. Year 1 (Bac+4)**
3-month internship
Exploring analogical proportions to align Knowledge Graphs
Co-supervised with Miguel Couceiro and Esteban Marquer

Oct. 2022 – Jun. 2023 **Omar Abedelkader, Juba Ait-Abdelmalek, Ekaterina Kozachenko, M.Sc. Year 1 (Bac+4)**
8-month research project
Exploring analogies for automatic semantic annotation taking into account context
Co-supervised with Stergos Afantenos, Miguel Couceiro, and Esteban Marquer

Oct. 2022 – Jun. 2023 **Adrien Chassaing-Monjou, Chadi Hani, M.Sc. Year 1 (Bac+4)**
8-month research project
Exploring analogical proportions between and within Knowledge Graphs
Co-supervised with Miguel Couceiro and Esteban Marquer

Sept. 2022 – Feb. 2023 **Li-Fan Zhao, M.Sc. Year 2 (Bac+5)**
5-month research project
Knowledge extraction from tabular data in Electronic Health Records
Co-supervised with Adrien Coulet and Bastien Rance

Mar. 2022 – Sept. 2022 **Lucas Jarnac, M.Sc. Year 2 (Bac+5)**
6-month internship
Creation of an Enterprise Knowledge Graph: bootstrapping and knowledge reconciliation
Publication: [27]

Jun. 2019 – Aug. 2019 **Melvin Moreau, B.Sc. Year 2 (IUT Bac+2)**
2-month internship
Updating PGxLOD with the newest versions of the integrated Linked Open Data sets
Co-supervised with Amedeo Napoli

Jan. 2018 – May 2018 **Benoît Frapiccini, Gabi Portelli, M.Sc. Year 1 (Bac+4)**
5-month research project
Extracting drug side effects from clinical annotations of PharmGKB to refine PGxLOD
Co-supervised with Adrien Coulet

Jun. 2017 – Aug. 2017 **Graziella Husson, M.Sc. Year 1 (Bac+4)**
3-month internship
Refining PGxLOD (integrating Linked Open Data sets, transforming PharmGKB data into RDF)
Co-supervised with Adrien Coulet
Publication: [7]

## Post-doc researchers (1)

Jun. 2026 – Aug. 2029 **Minh Hoang Dang**
Natural language based secure federated query building (SAFE-KG project)
Co-supervised with Fabien Gandon, Catherine Faron, and Franck Michel

## Engineers (3)

Nov. 2026 – Nov. 2027 **Tom Toupence**
Construction of a knowledge graph and implementation of a GraphRAG system
Co-supervised with Catherine Faron

Nov. 2024 – Sept. 2025 **Ndeye-Emilie Mbengue**
Development of the KGPrune web application (AT2TA project)
Co-supervised with Miguel Couceiro
Publication: [24]

Jan. 2024 – Sept. 2024 **Cherif Hassan Nousradine**
Development of the KGPrune web application (AT2TA project)
Co-supervised with Miguel Couceiro
Publication: [41, 65]

## Available software and data

- PGxLOD, a knowledge graph to integrate pharmacogenomic knowledge
  `https://pgxlod.loria.fr`
- PGxO, an ontology to represent pharmacogenomic knowledge
  `https://pgxo.loria.fr`
- YAGO4-LP, link prediction datasets based on the YAGO4 knowledge graph
  `https://doi.org/10.5281/zenodo.10677792`
- DB100k+, YAGO3-10+ and NELL-995+: semantically enriched datasets for link prediction
  `https://doi.org/10.5281/zenodo.17438317`
- Wikidata Thematic Subgraph Selection, datasets to train and evaluate algorithms to select thematic subgraphs of interest in a large knowledge graph from seed entities of interest
  `https://doi.org/10.5281/zenodo.8091583`
- Wikidata Thematic Subgraph Selection - An Artwork Dataset for experiments to select thematic subgraphs of interest in a large knowledge graph from seed entities of interest
  `https://doi.org/10.5281/zenodo.15600971`
- Pharmacogenomics datasets for Ontology Matching, datasets based on the PGxLOD knowledge graph to evaluate Ontology Matching algorithms on a structure-based instance matching task from the domain of pharmacogenomics
  `https://doi.org/10.5281/zenodo.8419361`
- analogical-pruning, an analogy-based pruning model for subgraph selection
  `https://github.com/Orange-OpenSource/analogical-pruning`
- PyGraft, a Python tool to generate synthetic schemas and knowledge graphs
  `https://github.com/nicolas-hbt/pygraft`
- KGPrune, an API and Web application to extract subgraphs from Wikidata around user-input entities of interest while avoiding topical drift
  `https://kgprune.loria.fr`
- `graph-seeder`, a Python package to extract, densify, and analyze subgraphs from knowledge graphs
  `https://github.com/Wimmics/graphseeder` `https://pypi.org/project/graph-seeder/`
- Code of other experiments on GitHub
  `https://github.com/pmonnin`

## Publications

Rankings of journals and conferences are provided for reference and are respectively drawn from the Scimago Journal Rank and the CORE Conference Portal. All publications can be accessed through my webpage: `https://pmonnin.github.io/publications.html`. An asterisk (*) denotes co-authors who contributed equally.

### International journals

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[58] Emmanuel Bresso*, **Pierre Monnin***, Cédric Bousquet, François-Élie Calvier, Ndeye Coumba Ndiaye, Nadine Petitpain, Malika Smaïl-Tabbone, and Adrien Coulet. *Investigating ADR mechanisms with Explainable AI: a feasibility study with knowledge graph mining*. JOBIM 2022 - Journées Ouvertes en Biologie, Informatique et Mathématiques, Proceedings - Keynotes, Contributed talks, Mini-Symposia, Rennes, France, July 05-08, 2022. URL: https://hal.inria.fr/hal-03754898.

**Abstracts of oral presentations in national workshops**

[59] Yoan Chabot, Thomas Labbé, and **Pierre Monnin**. *DAGOBAH : Des outils pour l'interprétation automatique de données tabulaires*. SemWeb.Pro 2021, Remote, December 9, 2021. URL: https://2021.semweb.pro/presentations/dagobah/.

**Thesis**

[60] **Pierre Monnin**. "Matching and mining in knowledge graphs of the Web of data - Applications in pharmacogenomics. (Appariement et fouille dans les graphes de connaissances du Web des données - Applications en pharmacogénomique)". PhD thesis. University of Lorraine, Nancy, France, 2020. URL: https://tel.archives-ouvertes.fr/tel-03122326.

**Editorships**

[61] Miguel Couceiro, Esteban Marquer, **Pierre Monnin**, and Pierre-Alexandre Murena, eds. *Special issue on analogies: from mathematical foundations to applications and interactions with ML and AI*. Vol. 93. Annals of Mathematics and Artificial Intelligence 2. 2025, pp. 233–235. DOI: 10.1007/S10472-024-09961-7. URL: https://inria.hal.science/hal-04858719.

[62] Miguel Couceiro, Erkko Lehtonen, and **Pierre Monnin**, eds. *Special issue to the memory of Maurice Pouzet*. Vol. 23 (2). Discrete Mathematics and Theoretical Computer Science. 2023. URL: https://inria.hal.science/hal-04860979.

[63] Ulrike Sattler, Aidan Hogan, Maria Keet, Valentina Presutti, João Paulo A. Almeida, Hideaki Takeda, **Pierre Monnin**, Giuseppe Pirrò, and Claudia d'Amato, eds. *The Semantic Web - ISWC 2022 - 21st International Semantic Web Conference, ISWC 2022, Virtual Event, October 23-27, 2022, Proceedings*. Lecture Notes in Computer Science. Springer, 2022. DOI: 10.1007/978-3-031-19433-7.

[64] Miguel Couceiro, **Pierre Monnin**, and Amedeo Napoli, eds. *Proceedings of the First International Conference "Algebras, graphs and ordered sets", ALGOS 2020, Online (originally planned in Nancy, France), August 26-28, 2020*. Vol. 2925. CEUR Workshop Proceedings. CEUR-WS.org, 2021. URL: http://ceur-ws.org/Vol-2925.

**Popularization articles**

[65] **Pierre Monnin**, Cherif-Hassan Nousradine, Lucas Jarnac, Laurel Zuckerman, and Miguel Couceiro. "KGPrune : une application Web pour extraire des sous-graphes d'intérêt de Wikidata par élagage analogique". In: *1024 : Bulletin de la Société Informatique de France* 24 (Nov. 2024), pp. 177–188. DOI: 10.48556/SIF.1024.24.177.

[66] **Pierre Monnin** and Adrien Coulet. "Projet ANR (2016-2021) " PractiKPharma " : extraction, comparaison et découverte de connaissances en pharmacogénomique". In: *1024 : Bulletin de la Société Informatique de France* 19 (Apr. 2022), pp. 109–120. DOI: 10.48556/sif.1024.19.109.

[67] **Pierre Monnin** and Adrien Coulet. "Étudier les mécanismes des effets indésirables des médicaments avec l'IA explicable : expériences avec la fouille de graphes de connaissances". In: *Bulletin de l'Association Française pour l'Intelligence Artificielle* 116 (Apr. 2022), pp. 37–40. URL: https://hal.inria.fr/hal-03669065.

**Popularization oral presentations**

[68] **Pierre Monnin**. "Using Wikidata to bootstrap a new knowledge graph: selecting relevant entities using analogical pruning". Wikidata Modelling Days 2023, Virtual Event, November 30, December 1-2, 2023. URL: https://www.youtube.com/watch?v=euXs8Jz4aeg.

**Preprints**

[69] Pierre Maillot, Catherine Faron, Fabien Gandon, and Franck Michel **Pierre Monnin**. "Eat your own KR: a KR-based approach to index Semantic Web Endpoints and Knowledge Graphs". 2025. URL: https://doi.org/10.48550/arXiv.2508.08713.

[70] Nicolas Hubert, **Pierre Monnin**, and Heiko Paulheim. "Beyond Transduction: A Survey on Inductive, Few Shot, and Zero Shot Link Prediction in Knowledge Graphs". 2023. URL: https://arxiv.org/abs/2312.04997.

## Invited talks

### International workshops

- **Pierre Monnin**. Discovering alignment relations with Graph Convolutional Networks: a biomedical case study. ROCED workshop at KGC 2022, May 2, 2022, Remote.

### National conferences

- **Pierre Monnin**. Matching and mining in knowledge graphs of the Web of data - Applications in pharmacogenomics. EGC 2022 (best thesis award), January 24-28, 2022, Blois, France.

### National workshops

- **Pierre Monnin**. Matching and mining in knowledge graphs of the Web of data - Applications in pharmacogenomics. DECADE workshop at PFIA 2022, June 28, 2022, Saint-Etienne, France.
- Yoan Chabot, **Pierre Monnin**. DAGOBAH : Activités de recherche Orange autour de l'annotation sémantique de données tabulaires. TextMine'21 workshop at EGC 2021, January 26, 2021, Montpellier, France, Remote.

### Seminars, symposiums, and junior conferences

- **Pierre Monnin**. Neurosymbolic AI for and with knowledge graphs. ENFIELD Adaptative AI Webinar, June 22, 2026, Online.
- **Pierre Monnin**. Rethinking Link Prediction: Instability, Semantic Awareness, and Neuro-Symbolic Approaches. 3IA Côte d'Azur Doctoral & Postdoctoral Seminar, June 12, 2026, Sophia-Antipolis, France.
- **Pierre Monnin**. Rethinking Link Prediction: Instability, Semantic Awareness, and Neuro-Symbolic Approaches. DaKoDAC seminar, HeKA team, April 17, 2026, Paris, France.
- **Pierre Monnin**. Aligning complex units in knowledge graphs – Symbolic and neuro-symbolic approaches for pharmacogenomics. SIMDAC 2025, MOBE, Université d'Orléans, June 24, 2025, Orléans, France.
- **Pierre Monnin**. Neuro-symbolic approaches for the knowledge graph lifecycle. DIG team seminar, LTCI, Télécom Paris, March 18, 2025, Paris, France.
- **Pierre Monnin**. The Schema Strikes Back: Refining Knowledge Graphs with Neuro-Symbolic AI. Università degli Studi di Bari Aldo Moro – Seminar of the PhD Program in Computer Science and Mathematics, January 27, 2025, Bari, Italy.
- **Pierre Monnin**. Neuro-symbolic approaches for the knowledge graph lifecycle. INESC-ID, Universidade de Lisboa – Seminar of the Language & Speech working group, November 22, 2024, Lisbon, Portugal.
- **Pierre Monnin**. Aligning complex units in knowledge graphs - Symbolic and neuro-symbolic approaches for pharmacogenomics. ANR AGGREEY meeting, June 4, 2024, Sophia-Antipolis, France.
- **Pierre Monnin**. Neuro-symbolic approaches for the knowledge graph lifecycle. Università degli Studi di Bari Aldo Moro – Seminar of the PhD Program in Computer Science and Mathematics, January 30, 2024, Bari, Italy.
- **Pierre Monnin**. Neuro-symbolic approaches for the knowledge graph lifecycle. MALOTEC seminar, LORIA, December 18, 2023, Nancy, France.
- **Pierre Monnin**. Knowledge graphs and analogical reasoning: from the example of zero-shot KG bootstrapping to perspectives. Workshop on analogies : from learning to explainability (GT MHyIA & EXPLICON – GdR RADIA), November 27–28, 2023, Arras, France.
- **Pierre Monnin**. Knowledge graph construction, refinement, and applications, with a neuro-symbolic perspective. GT MHyIA meeting, March 29, 2023, Lens, France.
- **Pierre Monnin**. Knowledge graph construction, refinement, and applications, with a neuro-symbolic perspective. WIMMICS seminar, March 23, 2023, Sophia-Antipolis, France.
- **Pierre Monnin**. Knowledge graphs: construction, matching, and applications. CRIL seminar, January 12, 2023, Lens, France.
- **Pierre Monnin**. Knowledge graphs: construction, matching, and mining. 7th Junior Conference on Data Science and Engineering (JDSE 2022), September 15, 2022, Palaiseau, France.

- **Pierre Monnin**. Matching and mining in knowledge graphs of the Web of data - Applications in pharmacogenomics. 4th GdR MaDICS symposium, July 11, 2022, Lyon, France.
- **Pierre Monnin**. DAGOBAH - An End-to-End Context-Free Semantic Annotation System for Tabular Data. MALOTEC seminar, LORIA, February 4, 2021, Nancy, France, Remote.

## Non-academic talks / popularization activities

- Ndeye-Emilie Mbengue, **Pierre Monnin**. Knowledge Graphs & Analogical Reasoning. Visit of students from EURECOM to the Centre Inria d'Université Côte d'Azur. November 5, 2025. Sophia-Antipolis, France (∼50 people)
- Ndeye-Emilie Mbengue, **Pierre Monnin**. Knowledge Graphs & Analogical Reasoning. Visit of students from ENS Lyon to the I3S Laboratory. November 5, 2025. Sophia-Antipolis, France (∼30 people)
- **Pierre Monnin**. Knowledge graphs: construction, refinement, and applications, with a neuro-symbolic perspective. Language & Cognition seminar, IDMC, Université de Lorraine, February 1, 2023, Nancy, France (∼30 people).
- Mathieu d'Aquin, Miguel Couceiro, **Pierre Monnin**. From machine learning to knowledge and back: A (de)tour over reasoning, analogies and knowledge graphs. Panel discussion. Cognitive Science and NLP forum of IDMC, Université de Lorraine, November 23, 2022, Nancy, France (∼30 people).
- **Pierre Monnin**. Presentation of my career. TELECOM Nancy career day, June 9, 2022, Nancy, France (∼75 people).
- **Pierre Monnin**. Valedictorian speech. TELECOM Nancy graduation ceremony, November 28, 2015, Nancy, France (∼300 people).

## Teaching

Classes with $^{(EN)}$ were taught in English. Other classes were taught in French.

## Summary

| Year | Class | Level | Lecture | Tutorial | Practical | Resp. |
|---|---|---|---|---|---|---|
| 2025 – 2026 | AI for Adult Education | M.Sc. Year 1 | 24 | | | × |
| | ML & Semantic Web | M.Sc. Year 2 | 2 | 3 | | × |
| | NoSQL Databases | M.Sc. Year 1 | 6 | 15 | | × |
| 2024 – 2025 | AI for Foreign Languages: Professional Approaches | M.Sc. Year 2 | 9 | 9 | | × |
| | AI for Applied Foreign Languages | M.Sc. Year 1 | 9 | 9 | | × |
| | AI for Adult Education | M.Sc. Year 1 | 24 | | | × |
| | AI for preparatory classes for higher ed. | B.Sc. Year 1 | 3 | | | × |
| | AI for manufacturing and service | B.Sc. Year 2 | | 6 | | |
| | ML & Semantic Web | M.Sc. Year 2 | 1 | 1.5 | | × |
| | NoSQL Databases | M.Sc. Year 1 | 6 | 15 | | × |
| 2023 – 2024 | AI for Applied Foreign Languages | M.Sc. Year 1 | 12 | 12 | | × |
| | AI for Adult Education | M.Sc. Year 1 | 24 | | | × |
| | AI for Marketing | B.Sc. Year 3 | 12 | | | × |
| | AI for preparatory classes for higher ed. | B.Sc. Year 1 | 3 | | | × |
| | ML & Semantic Web | M.Sc. Year 2 | 1 | 1.5 | | × |
| | XML technologies | M.Sc. Year 1 | | 12.5 | | |
| | Machine learning tutoring | M.Sc. Year 2 | | 10 | | × |
| | Artificial Intelligence | M.Sc. Year 2 | 10 | 8 | | × |
| 2022 – 2023 | Knowledge-based systems | M.Sc. Year 2 | 4 | 4 | | × |
| 2019 – 2020 | Introduction to GNU/Linux and Git[EN] | M.Sc. Year 2 | | 4 | | |
| | Ontologies[EN] | M.Sc. Year 2 | | 6 | | |
| | Algorithms and programming | M.Sc. Year 1 | | 15 | | |
| | Semantic Web[EN] | M.Sc. Year 1 | | 15 | | |
| | Computability and Complexity[EN] | M.Sc. Year 1 | | 10 | | |
| | Computational Complexity | B.Sc. Year 3 | 15 | 15 | | × |
| | Advanced Java Programming | B.Sc. Year 3 | | 20 | | |
| | Java Programming | B.Sc. Year 2 | | 20 | | |
| | Web Programming | B.Sc. Year 1 | | 14 | | |
| | Algorithmics | B.Sc. Year 1 | | 22 | | |
| | Tools and Digital Culture | B.Sc. Year 1 | | 20 | | |
| 2018 – 2019 | Artificial Intelligence | M.Sc. Year 2 | | 14 | 2 | |
| | Knowledge-Based Systems | M.Sc. Year 2 | 16 | 16 | | × |
| | NoSQL Databases | M.Sc. Year 2 | | | 20 | |
| | Compiler Implementation Project | M.Sc. Year 1 | | | 14 | |
| 2017 – 2018 | Artificial Intelligence | M.Sc. Year 2 | 2 | 14 | 2 | |
| | Knowledge-Based Systems | M.Sc. Year 2 | 16 | 16 | | × |
| | Compiler Implementation Project | M.Sc. Year 1 | | | 14 | |
| 2016 – 2017 | Theory of Compilation | M.Sc. Year 1 | | 20 | | |
| | Compiler Implementation Project | M.Sc. Year 1 | | | 14 | |
| | Computer Science Basics | B.Sc. Year 3 | | 18 | 16 | |
| 2015 – 2016 | Compiler Implementation Project | M.Sc. Year 1 | | | 14 | |
| | | **Total** | **199h** | **365.5h** | **96h** | |

A × in column "Responsibilities" (Resp.) means that I was in charge of the class (scheduling sessions, preparing subjects and handouts for lectures, tutorials, practicals, exams, and projects, coordination with other teachers, participation in examination boards). Grayed lines correspond to academic supervision of activities such as research projects or internships.

## Class descriptions

2024 – 2025 **Temporary lecturer – Université Côte d'Azur – Nice, France**

Introduction to AI for Adult Education – EFELIA Côte d'Azur

Introduction to Artificial Intelligence, Machine Learning, Deep Learning, training process, Neural Networks, evaluation, Natural Language Processing, associated tasks, impacts on society and environment

Machine Learning & Semantic Web – Polytech Nice Sophia

Introduction to machine learning with and for knowledge graph, knowledge graph embedding

models, the link prediction task, its evaluation, and GraphRAG approaches

NoSQL Databases – Polytech Nice Sophia
MongoDB, Elasticsearch, Redis, HBase, Neo4J, faiss

2024 – 2025 **Junior Fellow in AI – EFELIA Côte d'Azur – Nice, France**

AI for Foreign Languages: Professional Approaches – EFELIA Côte d'Azur
Focus on Natural Language Processing, tasks, (past to modern) models, limits, and applications in professional settings

Introduction to AI for Applied Foreign Languages – EFELIA Côte d'Azur
Introduction to Artificial Intelligence, Machine Learning, Deep Learning, training process, Neural Networks, evaluation, Natural Language Processing, associated tasks, impacts on society and environment

Introduction to AI for Adult Education – EFELIA Côte d'Azur
Introduction to Artificial Intelligence, Machine Learning, Deep Learning, training process, Neural Networks, evaluation, Natural Language Processing, associated tasks, impacts on society and environment

Introduction to AI for preparatory classes for higher ed. (namely, CPES) – EFELIA Côte d'Azur
Introduction to Artificial Intelligence, Machine Learning, Deep Learning, training process, Neural Networks, evaluation, associated tasks, impacts on society and environment

Introduction to AI for manufacturing and services (namely, BUT2 QLIO) – EFELIA Côte d'Azur
Introduction to Artificial Intelligence, Machine Learning, Deep Learning, training process, Neural Networks, evaluation, Generative AI / LLMs, associated tasks and challenges, with applications in manufacturing and services

Machine Learning & Semantic Web – Polytech Nice Sophia
Introduction to graph machine learning for knowledge graph refinement, some knowledge graph embedding models, link prediction, and evaluation

NoSQL Databases – Polytech Nice Sophia
MongoDB, Elasticsearch, Redis, HBase, Neo4J, faiss

2023 – 2024 **Junior Fellow in AI – EFELIA Côte d'Azur – Nice, France**

Introduction to AI for Applied Foreign Languages – EFELIA Côte d'Azur
Introduction to Artificial Intelligence, Machine Learning, Deep Learning, training process, Neural Networks, evaluation, Natural Language Processing, associated tasks, impacts on society and environment

Introduction to AI for Adult Education – EFELIA Côte d'Azur
Introduction to Artificial Intelligence, Machine Learning, Deep Learning, training process, Neural Networks, evaluation, Natural Language Processing, associated tasks, impacts on society and environment

Introduction to AI for Marketing (namely, BUT3 TC) – EFELIA Côte d'Azur
Introduction to Artificial Intelligence, Machine Learning, Deep Learning, training process, Neural Networks, evaluation, Natural Language Processing, image generation, associated tasks, impacts on society and environment

Introduction to AI for preparatory classes for higher ed. (namely, CPES) – EFELIA Côte d'Azur
Introduction to Artificial Intelligence, Machine Learning, Deep Learning, training process, Neural Networks, evaluation, associated tasks, impacts on society and environment

Machine Learning & Semantic Web – Polytech Nice Sophia
Introduction to graph machine learning for knowledge graph refinement, some knowledge graph embedding models, link prediction, and evaluation

XML technologies – Polytech Nice Sophia
XML language, DTD, XQuery, XSLT

Machine learning tutoring – Polytech Nice Sophia
Basics of Machine Learning, supervised and unsupervised learning, neural networks (MLP, CNN), training process, evaluation

2023 – 2024 **Temporary lecturer – TELECOM Nancy – Nancy, France**

Artificial Intelligence
Artificial Intelligence, supervised and unsupervised learning (*e.g.*, decision trees, neural networks, backpropagation algorithm, multilayer perceptron, K-means, Kohonen maps), Markov decision processes, knowledge graphs and the Semantic Web (RDF, ontologies, OWL reasoning, SPARQL), introduction to advanced AI techniques (e.g., CNN, GNN), applications & limits of AI, eXplainable AI, neuro-symbolic AI

2022 – 2023 **Temporary lecturer – TELECOM Nancy – Nancy, France**

Knowledge-based systems
Semantic Web (RDF, ontologies, OWL reasoning, SPARQL)

2019 – 2020 **Temporary teaching and research fellow – IDMC – Nancy, France**

Introduction to GNU/Linux and Git(EN)
Basics of Linux and Git commands

Ontologies(EN)
Description Logics, SPARQL

Algorithms and Programming
Introduction to programming in Python (functional and object-oriented programming)

Semantic Web(EN)
First-order logic, Description Logics, Formal Concept Analysis, itemsets, association rules, Knowledge Graphs, Semantic Web (RDF, SPARQL) and applications

Computability and Complexity(EN)
Turing machines, uncomputability, undecidability, complexity (*e.g.*, big O notation)

Computational Complexity
Best, worst, and average case complexity, sorting algorithms, recursion, de-recursion, randomized algorithms, greedy algorithms, dynamic programming, backtracking, data structures (stacks, queues, heaps, graphs)

Advanced Java Programming
Advanced concepts of the Java language (collections, threads, User Interface)

Java Programming
Basic concepts of the Java language and Object Oriented Programming

Web Programming
HTML, CSS, and an introduction to JavaScript

Algorithmics
Introduction to algorithms (variables, assignments, conditions, loops, subalgorithms)

Tools and Digital Culture
Proficient use of Word, Excel, and Internet Research

2016 – 2019 **Teaching assistant – TELECOM Nancy – Nancy, France**

Artificial Intelligence
Search algorithms (*e.g.*, A*, alpha-beta pruning), genetic algorithms, constraint satisfaction problems, Markov decision processes, supervised and unsupervised learning (*e.g.*, decision trees, neural networks, backpropagation algorithm, multilayer perceptron, K-means, Kohonen map)

Knowledge-Based Systems
Semantic Web (RDF, ontologies, OWL, SPARQL), Prolog

NoSQL Databases
MongoDB, Elasticsearch, Redis, HBase, Neo4J

Theory of Compilation
LR(0), SLR(1), LR(1) and LALR(1) parsers, Symbol Tables, Abstract Syntax Trees, assembly code generation

Compiler Implementation Project
Implementation of a compiler with ANTLR, from grammar definition to assembly code generation

Computer Science Basics
Sorting algorithms, complexity, recursivity, back-tracking, Hoare logic

2015 – 2016 **Temporary lecturer – TELECOM Nancy – Nancy, France**

Compiler Implementation Project

Implementation of a compiler with ANTLR, from grammar definition to assembly code generation